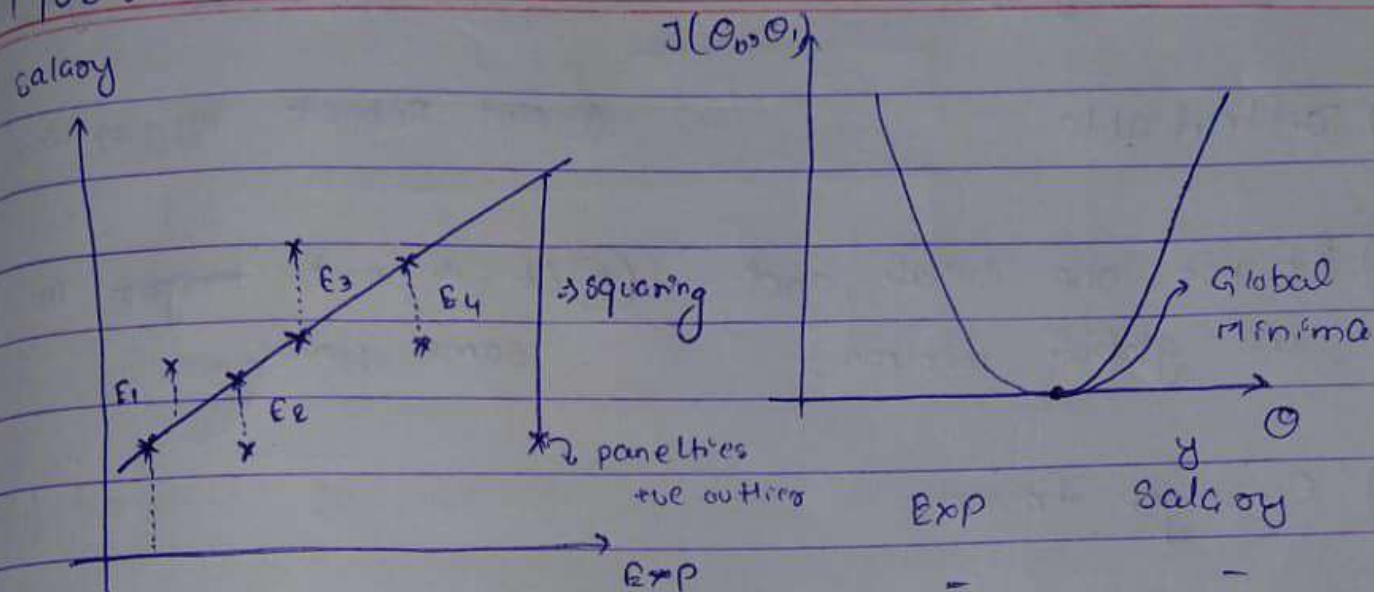


MSE, MAE, RMSE [cost function] → Performance Metrics

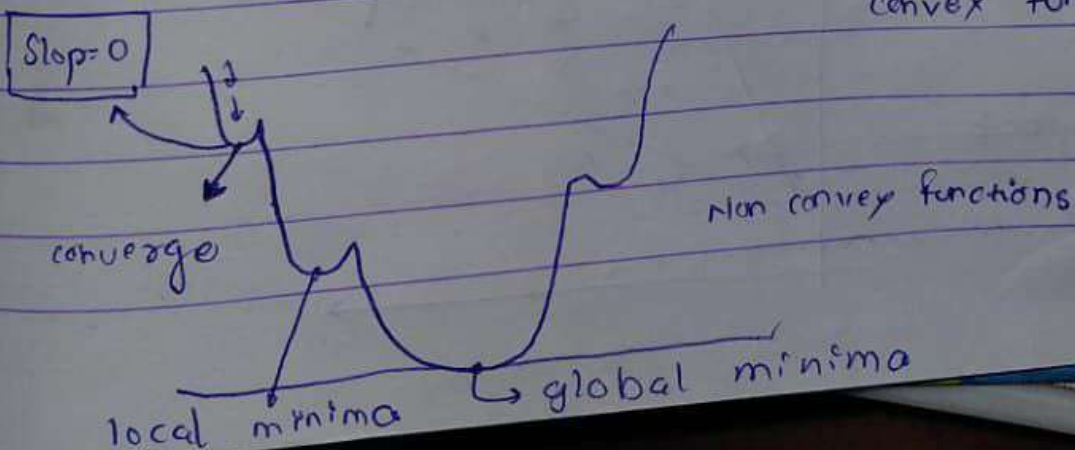
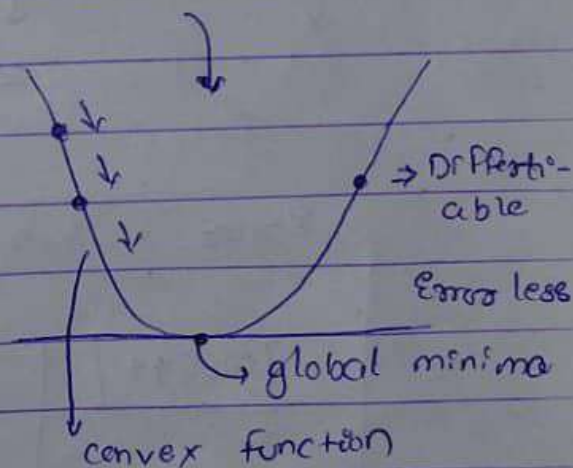


$$MSE = \sum_{i=1}^n \left(\frac{y - \hat{y}}{n} \right)^2 \Rightarrow \text{cost function} \downarrow -$$

Quadratic Equation

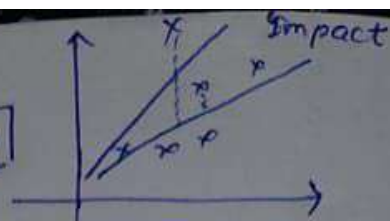
$$ax^2 + bx + c$$

$$(a-b)^2 = a^2 - 2ab + b^2$$



MSE

MSE



Disadvantage

Advantage

⊕ Not Robust to outliers

1) Differentiable

2) It has one local and one global minima.

⊖ It is not ~~linear~~ in same unit.

3) Converges faster

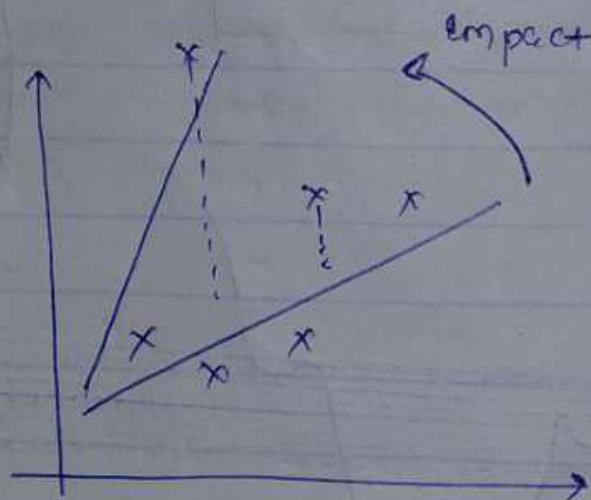
$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Salary (lacs) $\downarrow$ y

$(y_i - \hat{y}_i)^2$ (lacs)² (MSE)

$$Error = 2.5 \rightarrow (lacs)^2 \ll$$

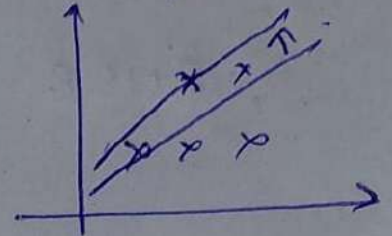
MSE



∴ Huge Impact

① Mean Absolute Error (MAE) 64 MAE ↑

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - \hat{y}|$$



Advantage

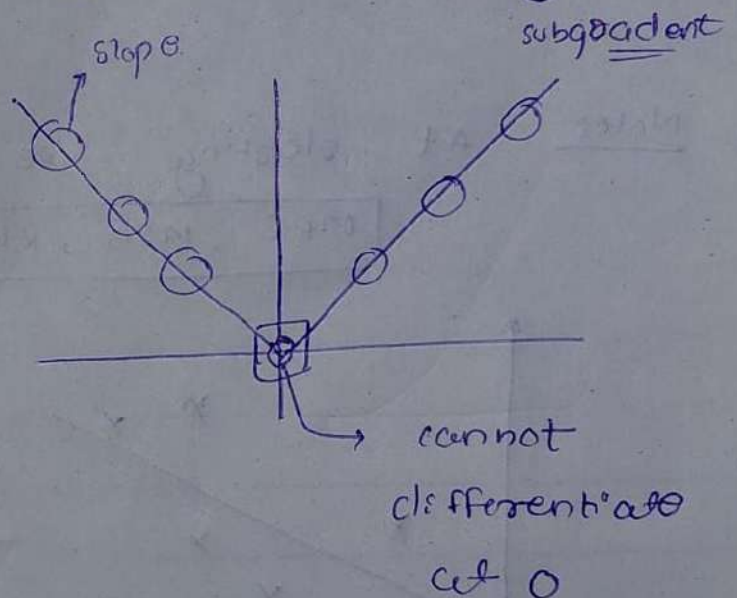
1) Robust to outliers

2) It will be on the same unit

Disadvantage

* convergence usually take more time. Optimization is a complex task

* Time consuming



③ RMSE (Root Mean Squared Error)

$$RMSE = \sqrt{MSE}$$

$$= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



Advantage

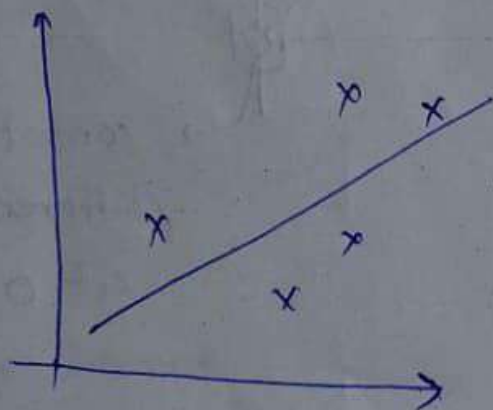
- 1) same unit
- 2) differentiable

Disadvantage

- Not Robust to outliers

Note: At starting we focus all of these Performance metrics

MAE, MSE, RMSE



- MAE
- MSE
- RMSE
- R^2
- Adjusted R^2

Interview :-

MSE vs MAE vs RMSE
 R^2 vs Adjusted R^2